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Implementation and transition to post-quantum cryptography of the Minimal IKE protocol

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Asymmetric cryptography

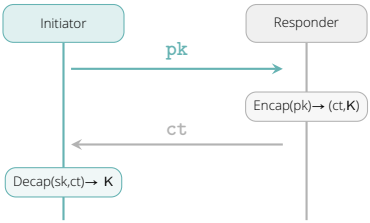
The building blocks of security protocols are **asymmetric cryptographic primitives** .

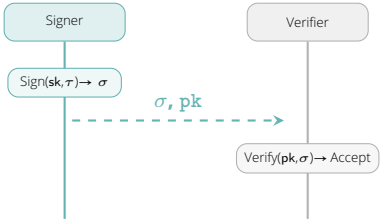
Key Encapsulation Mechanisms

Establish a shared secret **K** between peers.
Guarantees: **confidentiality**

Digital Signature Schemes

Authenticate peers by signing transcript τ.
Guarantees: **authentication**

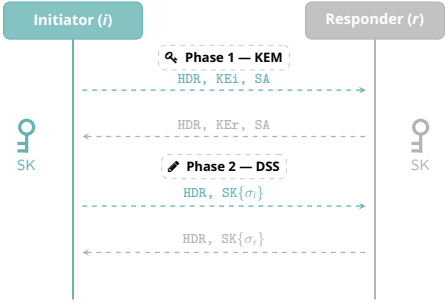




IPsec and IKE

IKEv2 is the standard protocol used by **IPsec VPNs** to establish secure communication channels.
Main exchanges of the protocol:

- `IKE_SA_INIT`: establishes secret
- `IKE_AUTH`: authenticates communicating peers



⇒ Simple, but over time many additional features were introudce to extend its functionality.
Making it unusable in constrained environments.

Post-Quantum Transition Challenges

- It is well known that classical asymmetric cryptography is vulnerable to attacks based on quantum computers.
- The time has now come to replace it with post-quantum cryptography, which has already been standardized.
- However, this has some impact on the protocol.

Communication cost increases:

PQ primitives introduce significantly larger public keys and ciphertexts:

	pk size	ct size
`X25519`	32 B	32 B
`ML-KEM-512`	800 B	768 B

Structural problem:

`INIT` messages may exceed the MTU. Fragmentation is the natural fix, **but**:

- peers are *unauthenticated*
- Fragmentation ⇒ **DoS exposure**
- Mitigations add *protocol complexity*

⇒ PQ security comes at the cost of **increased complexity**.

Open Challenges & Contributions

Exists a way to make IKE lightweight?

Yes, RFC 7815 defines *Minimal IKE*.
But: no efficient open-source implementation exists.

Efficient implementation of Minimal IKE (RFC 7815) — **Colibri**.

Is it quantum-safe?

No, RFC 7815 predates the NIST PQC competition.
And: in constrained networks, the PQC transition introduces new *communication* overhead that may be more critical than computational overhead.

First post-quantum variant of Minimal IKE.

Minimal IKE

Minimal IKE redesigns IKEv2 for constrained devices by:

- **simplifying the core protocol**
- and **removing unnecessary extensions**.

Core protocol — simplified

- Only INIT and AUTH exchanges
- Limited cipher suite
- PSK authentication only
- Initiator-only role
- Static message parsing

⊘ Extensions — removed

- Fragmentation (RFC 7383)
- Re-keying (`CREATE_CHILD_SA`)
- MOBIKE (RFC 4555)
- EAP, notification payloads
- *Other optional extensions*

⇒ A leaner protocol design that retains the **essential security guarantees** of IKEv2.

Implementing Minimal IKE

Only a proof-of-concept implementation in Python was available.
We implemented the protocol **from scratch in C**, with the following design goals:

Design Goals

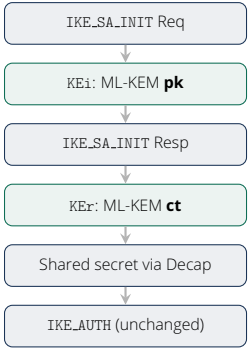
- *Minimal memory footprint* — no dynamic allocation, fixed-size buffers
- *Low computational overhead* — static configuration, no runtime negotiation
- *Small, self-contained code base* — minimal dependencies, no unnecessary abstractions
- *Portability* — clean separation between data structures and wire representation

⇒ Every design choice targets one goal: **keep it as small and predictable as possible**.

Post-Quantum Extension: ML-KEM into IKE

Specified in `draft-ietf-ipsecme-mlkem-ikev2`.
ML-KEM (FIPS 203) is used as a **drop-in replacement** for X25519 in the `KE` payload.

- Initiator sends **ML-KEM public key** in `KEi`
- Responder replies with **ciphertext** in `KEr`
- Shared secret derived via decapsulation
- AUTH exchange **unchanged**



⇒ PSK + ML-KEM-512 achieves **NIST security level 1** against a Quantum computer.
No added protocol complexity.

Experimental Setup and Benchmarking Methodology

Benchmarking goal

- Quantitatively assess `Colibri`'s lightness advantage vs. full `IKEv2` implementations
- Evaluate performance cost of replacing `X25519` with `ML-KEM-512`

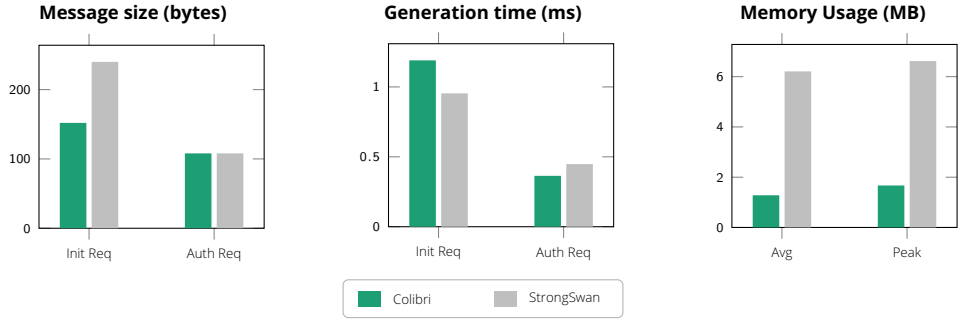
Comparison baseline: StrongSwan

- Industry-standard IKEv2 implementation
- **Minimised** configuration: non-core payloads disabled, only essential modules loaded
- Same cipher suite and authentication method for both
- All primitives sourced from **OpenSSL** for a fair baseline

Results: Pre-quantum configuration

Hardware details: CPU: `Ryzen 7 PRO 8840` @ 3.3 GHz • RAM: 16 GB

Software details: Library: `OpenSSL 3.6.2` • Threads: 1 vs 6 • OS: `Arch Linux`

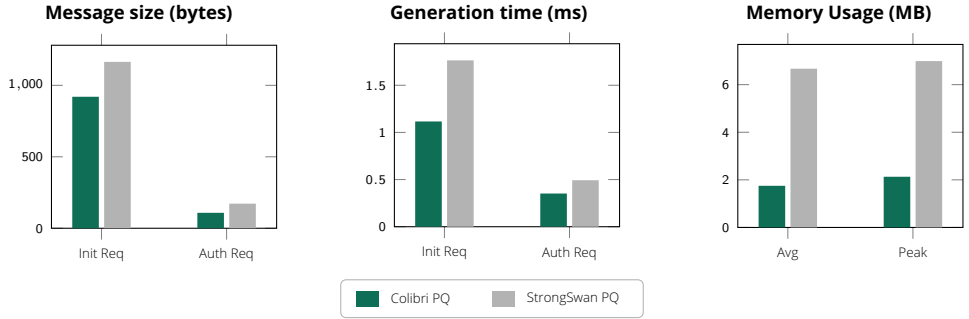


⇒ Slightly slower generation time, but **3.8× less memory**.

Results: Post-quantum configuration

Hardware details: CPU: `Ryzen 7 PRO 8840` @ 3.3 GHz • RAM: 16 GB •

Software details: Library: `OpenSSL 3.6.2` • Threads: 1 vs 6 • OS: `Arch Linux` • PQC: `liboqs`



⇒ Colibri **retains its advantages over StrongSwan even with ML-KEM**:
confirming that the PQ transition does not compromise the lightweight nature of Minimal IKE.

Key Findings

- Colibri is **significantly lighter** than a minimised StrongSwan — the feature drop directly translates into reduced memory footprint and code size.
- Switching from X25519 to ML-KEM-512 incurs **minimal overhead**: encap/decap is fast (lattice arithmetic via NTT), only the `KE` payload grows, no extra round-trips required.
- The **advantages are retained** even in the post-quantum configuration — Colibri PQ remains lighter than StrongSwan PQ in both memory and message size.
- In constrained deployments, the feature drop may be an **acceptable trade-off** for significantly reduced complexity and resource usage.

Conclusions

- Minimal IKE + ML-KEM-512 achieves **NIST security level 1** against a quantum adversary
- The protocol remains **fast and lightweight** even after the PQ transition
- Colibri is interoperable: Minimal IKE initiator ↔ any full IKEv2 responder

Future works

- How to handle KEM material at higher security level
- Performance evaluation on **actual embedded hardware** (e.g. ARM Cortex-M)
- Explore Hybrid approaches (classical& PQ) for KEM
- `IND-CPA` security for the KEM
- Use DSS in `AUTH` phase with minimal overhead

Acknowledgements

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Thank You!

Questions?

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